

QGIS —
Design/Formulários



QField — Coleta móvel
(offline-first)



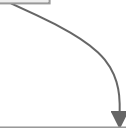
QFieldCloud —
Sincronização (Offline
Editing)



PostgreSQL/PostGIS
(Supabase) — Single
source of truth



Python: Pandas,
GeoPandas, NumPy,
SQLAlchemy — Motor de
cálculo



HTML5/CSS3/JS +
Leaflet.js + Chart.js —
Dashboard